

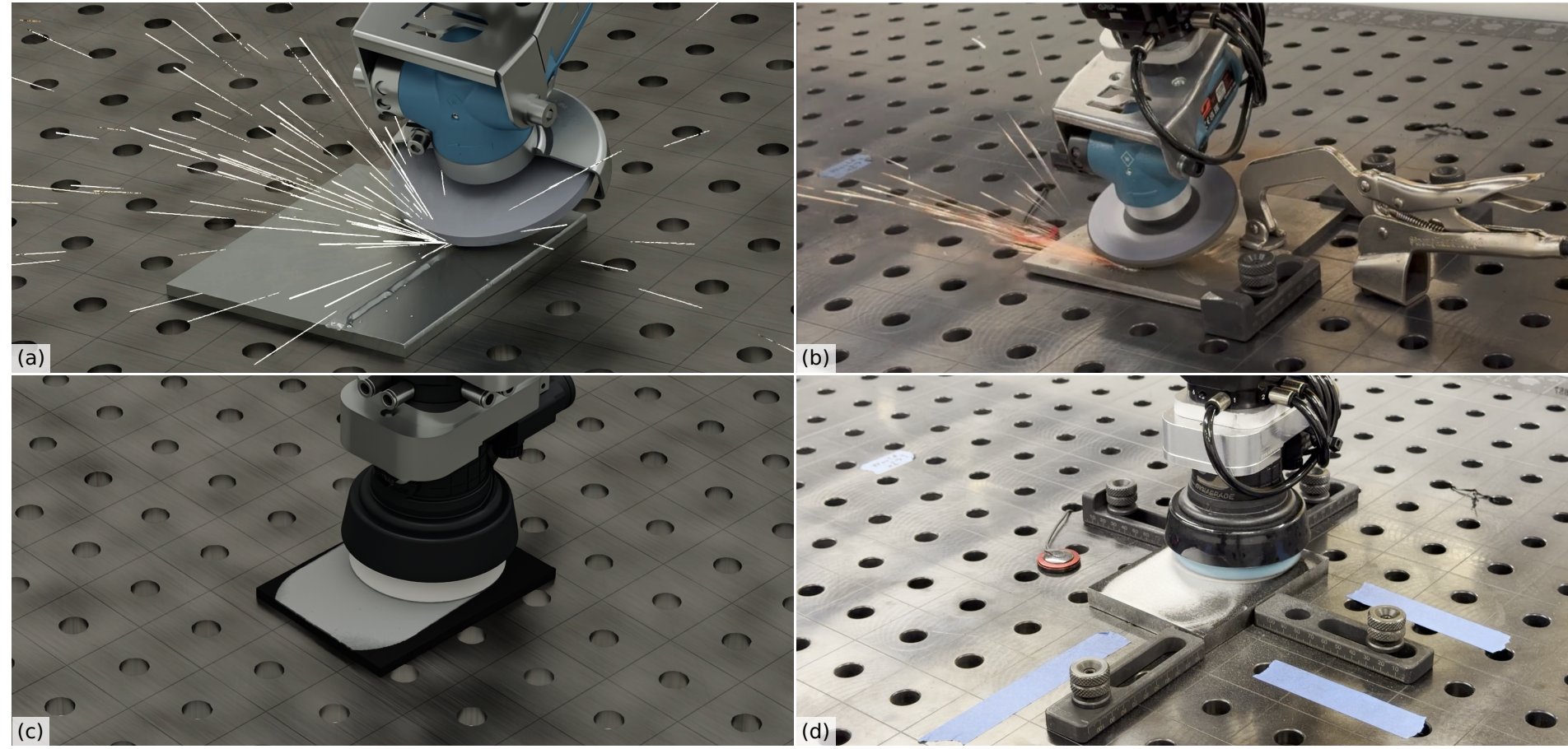
RoboFin3D: A Sim-to-Real Platform for Robotic Surface Finishing

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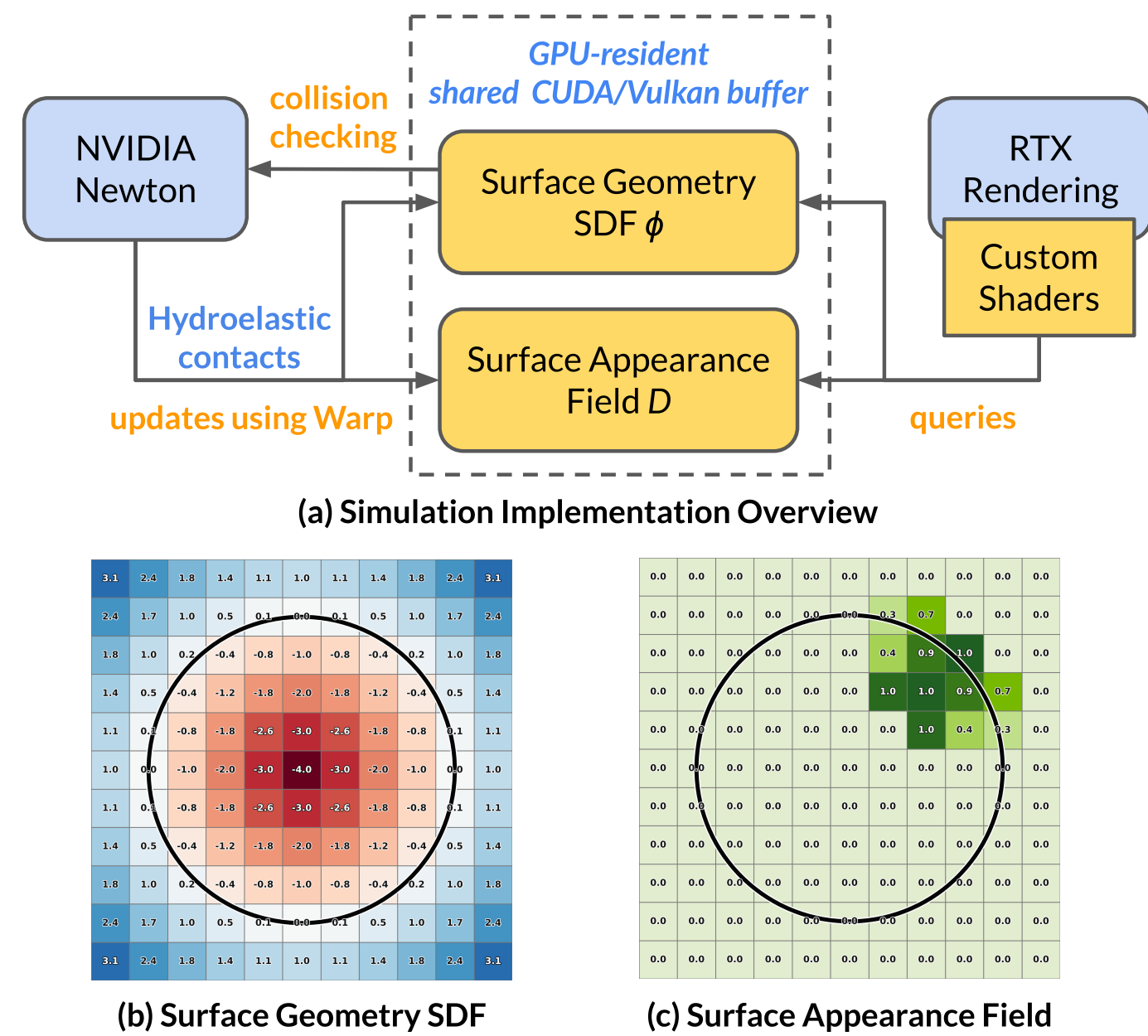
1 Motivation

Each grinding or sanding trial **permanently changes the workpiece**, so real experiments are costly and hard to reproduce. A simulator **restores any earlier state** at no material cost.



Simulation (a, c) and reality (b, d) of grinding with a rigid wheel (top) and sanding with a spinning pad (bottom).

2 Simulating an evolving workpiece

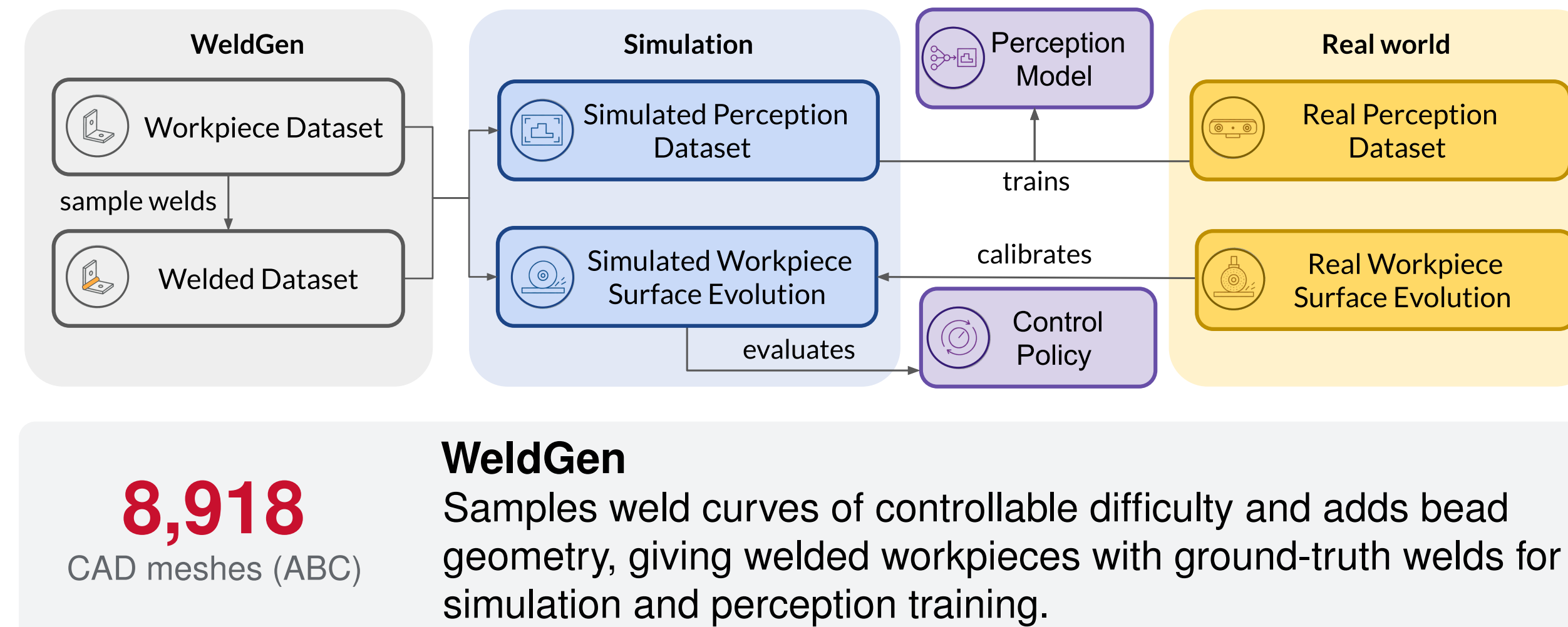


Isaac Sim + Newton. An SDF ϕ stores the geometry and a surface field D stores sanding progress in one shared GPU buffer. Contact, field updates, and rendering read the fields directly, with **no intermediate mesh extraction**.

Grinding $\partial_t \phi = r \|\nabla \phi\|$, $r = k \max(p - p_c, 0)$

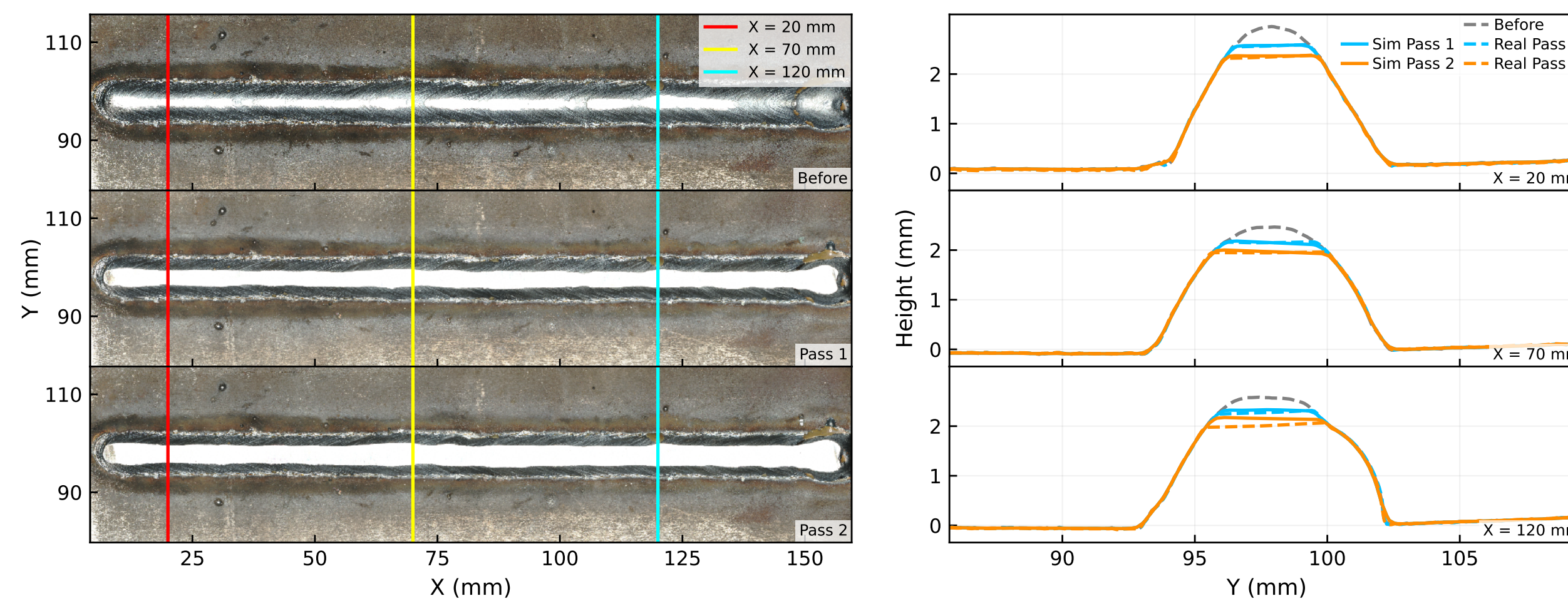
Sanding $D = \frac{K_P}{h_c} \int_0^t p v_{\text{slip}} d\tau$, coating removed at $D = 1$
 p : contact pressure; p_c : threshold pressure; v_{slip} : sliding speed

3 RoboFin3D workflow



4 Simulation vs. reality

For each process, a real workpiece is ground or sanded in **two passes** at a **15 N** force setpoint and scanned before and after each pass. The simulation uses the same initial scan, tool path, and force setpoint. **Pass 1 calibrates** the contact and removal parameters; **pass 2 is held out**, and the simulated pass 2 continues from the simulated pass 1.



Grinding. Scans (left) and height profiles (right). In simulation and reality, pass 1 turns the rounded crest into a plateau and pass 2 lowers it. The plateaus differ in height and slope, most after pass 2.

Grinding: removed volume

	Real	Sim	Error
Pass 1	139.1	111.0	20.2%
Pass 2	294.8	216.1	26.7%

Cumulative volume in mm³; error relative to Real. k is fitted to the pass 1 volume, then contact stiffness and p_c to height profiles.

Sanding: exposed region

0.81

IoU after pass 1

0.89

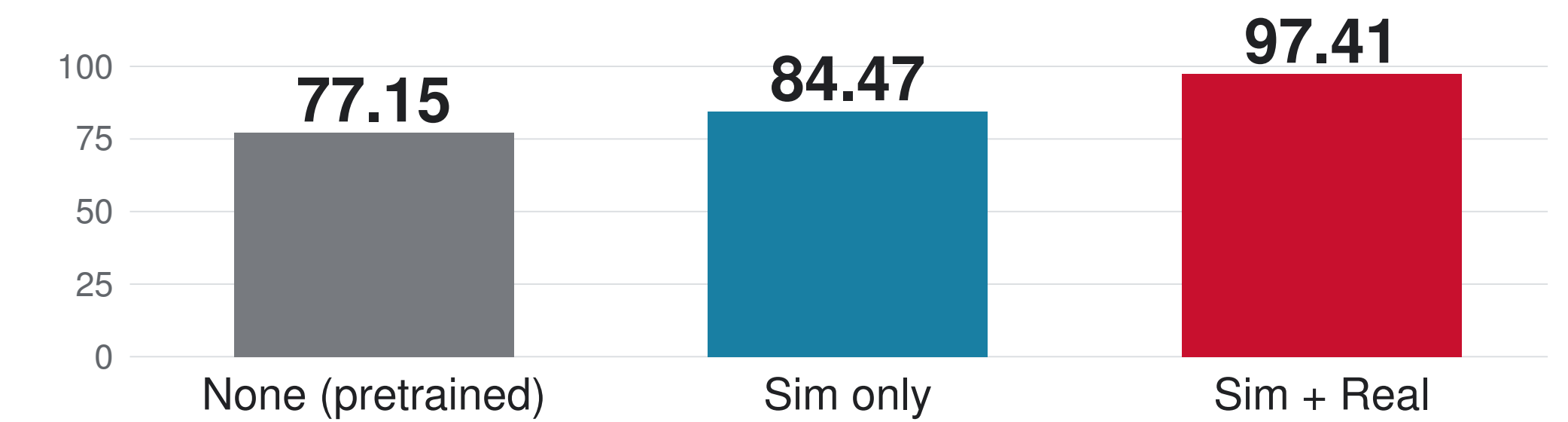
IoU after pass 2

Simulated $D \geq 1$ vs. exposed substrate in the scans; the simulated area is 16.5% and 11.2% larger. K_P/h_c is fitted to the pass 1 scan.

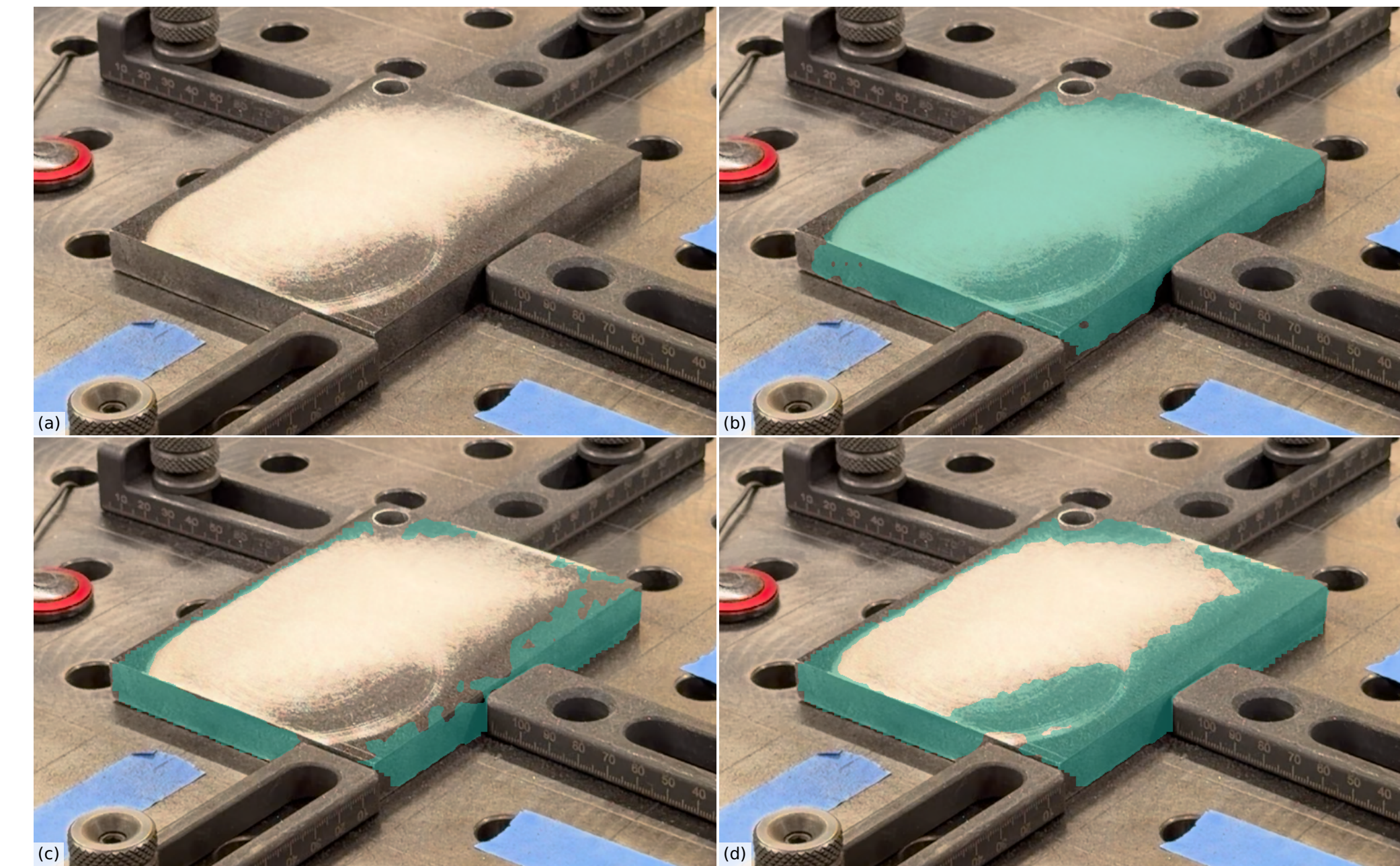
5 Sim-to-real perception

SAM2 learns to segment **unsanded regions** from simulated images that the simulator labels with **no manual annotation**.

IoU (%) on real test images, by fine-tuning data



SAM2.1 Tiny, frozen image encoder, ground-truth box prompts. 768 simulated images, split by placement into 512/128/128. Sim + Real adds labeled real images, about 20% of the simulated training set.



A real sanding image. (a) Input; (b) pre-trained SAM2; (c) Sim only; (d) Sim + Real. Teal overlays mark predicted unsanded regions.

Takeaway

RoboFin3D, a physics-based simulation platform for grinding and sanding, can lower the barrier of industrial robotic surface finishing, making experiments cheaper and more reproducible.

Future work

- Improving contact and removal models to better match real results
- Speeding up contact computation for real-time simulation
- Quantifying the sim-to-real gap across more workpieces and parameters
- Evaluating control policies in simulation before real deployment